1D Thermal Diffusion

Physical Fundamentals, Discretisation, and MATLAB Implementation

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1. Problem Geometry and Parameters

An aluminium bar with the following geometric and thermophysical properties is considered:

Table 1: Physical parameters of the problem

Quantity	Symbol	Value	Unit
Length	L	50	mm
Cross-sectional area	A	3×30	mm^2
Thermal diffusivity (Al)	α	98×10^{-6}	m^2/s
Thermal conductivity (Al)	k	237	$\mathrm{W}/(\mathrm{m\,K})$
Applied power at $x = 0$	q_{in}	25	W
Heat-sink temperature at $x = L$	T_{sink}	80	°C
Initial temperature	T_0	80	°C

The spatial domain is $x \in [0,\, L]$. The boundary conditions are:

- **Left boundary** ($x = 0$): prescribed heat flux **(Neumann)**;
- **Right boundary** ($x = L$): fixed temperature **(Dirichlet)**.

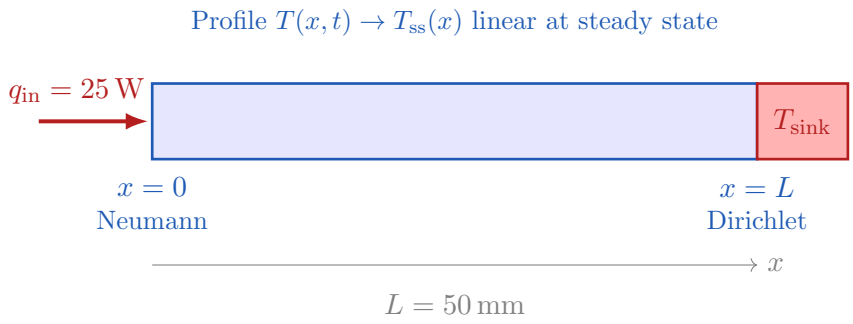


Figure 1: Domain schematic: bar with prescribed flux on the left and heat sink on the right.

2. Heat Equation (PDE)

2.1. General Differential Form

The heat diffusion equation in a homogeneous, isotropic medium, in one-dimensional form, is:

$$\boxed{\frac{\partial T}{\partial t}(x,t) = \alpha \, \frac{\partial^2 T}{\partial x^2}(x,t)} \tag{1}$$

where:

- $T(x,t)$ is the temperature distribution [°C] as a function of position x and time t;
- $\alpha = k/(\rho c_p)$ is the *thermal diffusivity* $[\mathrm{m}^2/\mathrm{s}]$, which measures the rate at which heat propagates through the material.

Equation (1) follows from the energy balance over an infinitesimal control volume $A\,\mathrm{d}x$, applying **Fourier's law**:

$$q = -kA\,\frac{\partial T}{\partial x} \tag{2}$$

which states that the heat flux is proportional to the temperature gradient and directed towards decreasing temperature.

2.2. Boundary Conditions

Left BC — Neumann (prescribed flux). The net heat flux leaving the bar in the $+x$ direction is zero at the boundary; the power q_{in} enters from the left edge:

$$-kA\,\frac{\partial T}{\partial x}\bigg|_{x=0} = q_{\text{in}} \quad \Longrightarrow \quad \frac{\partial T}{\partial x}\bigg|_{x=0} = -\frac{q_{\text{in}}}{kA} \tag{3}$$

Right BC — Dirichlet (prescribed temperature).

$$T(L,t) = T_{\text{sink}} = 80\,^{\circ}\text{C} \qquad \forall\, t > 0 \tag{4}$$

2.3. Initial Condition

The bar is initially at uniform temperature:

$$T(x,0) = T_0 = 80\,^{\circ}\text{C} \qquad \forall\, x \in [0, L] \tag{5}$$

3. Steady-State Solution

At steady state ($\partial T/\partial t = 0$), equation (1) reduces to:

$$\frac{\mathrm{d}^2 T}{\mathrm{d}x^2} = 0 \tag{6}$$

The general solution is a **linear** profile:

$$T_{\text{ss}}(x) = a\,x + b \tag{7}$$

Applying conditions (3) and (4):

$$a = \frac{\mathrm{d}T_{\text{ss}}}{\mathrm{d}x} = -\frac{q_{\text{in}}}{kA} \tag{8}$$

$$b = T_{\text{sink}} - a\,L \tag{9}$$

Substituting numerical values:

$$a = -\frac{25\,\text{W}}{237\,\text{W/(m\,K)} \times 9 \times 10^{-5}\,\text{m}^2} = -1170.96\,\text{K/m} \tag{10}$$

$$T_{\text{ss}}(0) = T_{\text{sink}} - a\,L = 80 - (-1170.96) \times 0.05 \approx 138.5\,^{\circ}\text{C} \tag{11}$$

The temperature at the hot end is therefore approximately 138.5 °C: the 58.5 °C drop over the 50 mm length is entirely due to the thermal resistance of the bar.

4. Discretisation: Finite Difference Method (FTCS)

4.1. Space–Time Grid

The continuous domain $[0, L] \times [0, T_{\text{final}}]$ is discretised into:

- N_x spatial nodes with step $\Delta x = L/(N_x - 1)$;
- time steps of size Δt (determined by the stability condition).

The notation $u_i^n \approx T(i\,\Delta x,\, n\,\Delta t)$ denotes the discrete temperature value.

4.2. FTCS Scheme for Interior Nodes

The **Forward-Time, Centered-Space (FTCS)** scheme is applied: forward difference for the time derivative, centered second difference for the spatial derivative:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \alpha\,\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} \tag{12}$$

Rearranging yields the explicit update formula:

$$\boxed{u_i^{n+1} = u_i^n + \lambda\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right)} \quad i = 2, \ldots, N_x - 1 \tag{13}$$

where λ is the **discrete Fourier number** (or CFL coefficient):

$$\lambda = \frac{\alpha\,\Delta t}{\Delta x^2} \tag{14}$$

4.3. Von Neumann Stability Condition

The FTCS scheme is *conditionally stable*. Von Neumann stability analysis requires:

$$\lambda \le \frac{1}{2} \tag{15}$$

In the simulation, $\lambda = 0.45$ is chosen (safety factor), giving:

$$\Delta t = \frac{0.45 \cdot \Delta x^2}{\alpha} \tag{16}$$

If $\lambda > 0.5$, the numerical solution *diverges*, regardless of the spatial grid quality.

5. Implementation of Boundary Conditions

5.1. Right BC — Dirichlet (node N_x)

The simplest condition: the rightmost node is pinned to the fixed value at every time step:

$$u_{N_x}^{n+1} = T_{\text{sink}} \qquad \forall\, n \tag{17}$$

5.2. Left BC — Neumann with Ghost Node (node $i = 1$)

The FTCS formula (13) requires node u_0^n (to the left of the boundary), which has no physical counterpart. A virtual **ghost node** u_0^n is introduced, determined by imposing the boundary derivative via a centred difference:

$$\left.\frac{\partial T}{\partial x}\right|_{x=0} \approx \frac{u_2^n - u_0^n}{2\,\Delta x} = -\frac{q_{\text{in}}}{kA} \tag{18}$$

Hence:

$$u_0^n = u_2^n + \frac{2\,\Delta x\, q_{\text{in}}}{kA} \tag{19}$$

Substituting (19) into the update equation for $i = 1$:

$$\begin{aligned} u_1^{n+1} &= u_1^n + \lambda\,(u_2^n - 2u_1^n + u_0^n) \\ &= u_1^n + \lambda\left(u_2^n - 2u_1^n + u_2^n + \frac{2\,\Delta x\, q_{\text{in}}}{kA}\right) \\ \boxed{u_1^{n+1} &= u_1^n + \lambda\left(2u_2^n - 2u_1^n + \frac{2\,\Delta x\, q_{\text{in}}}{kA}\right)} \end{aligned} \tag{20}$$

This approach guarantees **second-order** accuracy in the spatial approximation of the BC, in contrast to a simple forward difference which would be only first-order.

6. Summary of the Time-Stepping Loop

At each step $n \to n+1$ the simulation executes, in order:

1. **Update interior nodes** ($i = 2, \ldots, N_x - 1$) using formula (13);
2. **Left BC** — node $i = 1$ using formula (20) (ghost node);
3. **Right BC** — node $i = N_x$ pinned to T_{sink} via Dirichlet (17).

In MATLAB, steps 1–2 are implemented in vectorised form (no explicit loops), ensuring typical MATLAB efficiency:

```
% Interior nodes (FTCS)
u_new(2:end-1) = u(2:end-1) + lambda * ...
    (u(3:end) - 2*u(2:end-1) + u(1:end-2));

% Left BC: Neumann ghost node
u_ghost  = u(2) + 2 * dx * q_in / (k * A);
u_new(1) = u(1) + lambda * (u(2) - 2*u(1) + u_ghost);

```

```
% Right BC: Dirichlet
u_new(end) = T_sink;
```

Listing 1: Time-stepping loop kernel in MATLAB

7. Verification: Steady-State Error

At the end of the simulation, the numerical solution $u_i^{N_t}$ is compared against the analytical profile $T_{\mathrm{ss}}(x_i)$:

$$\varepsilon_{\max} = \max_i \left| u_i^{N_t} - T_{\mathrm{ss}}(x_i) \right| \tag{21}$$

With $N_x = 200$ and $\lambda = 0.45$, one expects $\varepsilon_{\max} \ll 0.1\,^{\circ}\mathrm{C}$, confirming the correctness of the discretisation and convergence to the expected physical steady state.

A. Nomenclature

$T(x,t)$	Temperature [°C]
α	Thermal diffusivity [m^2/s]
k	Thermal conductivity [W/m/K]
A	Cross-sectional area [m^2]
L	Bar length [m]
$q_{\mathbf{in}}$	Applied thermal power [W]
$T_{\mathbf{sink}}$	Heat-sink temperature [°C]
λ	Discrete Fourier number (CFL coefficient) [−]
Δx	Spatial step [m]
Δt	Time step [s]
u_i^n	Discrete temperature at node i, time instant n
u_0^n	Virtual ghost node for the Neumann BC